Gizem Toplu Tutay

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Education

The University of Texas at Austin **Austin, TX**
PhD in Operations Research and Industrial Engineering *Jan 2021 – Aug 2025 (expected)*
Focus: Stochastic programming. Power grid and healthcare infrastructure resilience.

Stanford University and DeepLearning.AI **Online**
Machine Learning Specialization *Certification, 2023*

Bogazici University **Istanbul, Turkey**
BS in Civil Engineering (Highest Honors, ranked 3rd) *Aug 2008 – May 2013*
Thesis: Investment decision on wind turbines and energy storage. MATLAB Programming. Result: an annual savings of 900 tons of carbon emissions and a $100,000 reduction in electricity costs.

Honors and Awards

- Student excellence scholarship for academic success (2008-2013)
- Graduation "Summa cum laude" (with highest honors) (2013)
- NSF award #1940308, CoPe EAGER: Addressing Human-Centric Decision-Making Challenges from Coastal Hazards via Integrated Geosciences Modeling and Stochastic Optimization (2021-2022)
- IC^2 Institute Seed Grant, Towards a Well Being Economy: Advancing Equity in Social, Economic, and Health Systems (2022-2023)

Publications

- G. Toplu-Tutay, J.J. Hasenbein, E. Kutanoglu (2022). "A Scenario-based Optimization Model for Long-term Healthcare Infrastructure Resilience against Flooding," Proceedings of the IISE Annual Conference & Expo.
- G. Toplu-Tutay, J.J. Hasenbein, E. Kutanoglu. "Impact of Power Outages Depends on Who Loses It: Equity-Informed Grid Resilience Planning via Stochastic Optimization," (Submitted).
- K.Y. Kim, G. Toplu-Tutay, J.J. Hasenbein, E. Kutanoglu. "A Stochastic Optimization Model for Patient Evacuation from Health Care Facilities during Hurricanes," (Submitted).

Invited Talks/Posters

- *A Scenario-based Optimization Model for Long-term Healthcare Infrastructure Resilience against Flooding*, IISE Conference, Seattle, May 2022
- *Equity-Informed Power Grid Resilience Planning via Stochastic Optimization*, INFORMS Annual Meeting, Indianapolis, October 2022
- Led annual seminars at CLK Energy Investment. Presented on electricity market developments to audience of over 50 people managing 25% of Turkey's total electricity demand (2015-2017).

Experience

The University of Texas at Austin **Austin, TX**
Graduate Research Assistant *Jan 2021 – present*
Bureau of Economic Geology - Comparing Electricity Options (CEO) project (ongoing)
- Multi-criteria decision making model to identify a reliable and resilient generation unit mix with minimum environmental impacts at the lowest cost under different energy transition scenarios

IC^2 Institute and NSF-funded previous research projects
- Equity-informed power grid resilience planning using stochastic optimization. Case study: ERCOT grid during hurricane-induced flooding.
- Stochastic MIP and a Greedy Heuristic for patient evacuation during extreme flooding.
- INFORMS and IISE presentations. Publications: 1 conference paper published, 2 journal papers submitted.

Teaching: Led "Engineering Statistics" recitations. Assisted in "Learning Structures" graduate course.

GISEMAI **Ankara, Turkey**
Founder *Jan 2018 – Dec 2020*
Led an e-commerce startup. Managed logistics, overseeing the entire process from manufacturing in China to warehousing and selling in the US. Applied inventory models for effective stock management.

CLK Energy Investment **Istanbul, Turkey**
Senior Strategic Planning Analyst *Jun 2015 – Nov 2017*
Led team of two. Worked with the departments across the enterprise. Influenced strategic decisions. Applied Time Series Forecasting in R programming for electricity demand prediction.

ENERJISA **Istanbul, Turkey**
Key Account Manager *Sep 2013 – Mar 2014*
Largest electricity generation company in Turkey

Notable Activities

- Member of INFORMS and IAEE
- Vice-Chair on the Tenant Advisory Board for University Housing and Dining: Serve as the voice of the residents and organize community-building events in university apartments(2023-present).
- Coordinated 4 conferences in Turkey, Azerbaijan (2), and Slovenia on behalf of International Association for Energy Economics (IAEE), Ohio, USA (2014-2016).
- Athletic Accomplishment: Silver Medalist in National Synchronized Ice Skating Competitions (2000-2008).